



Introduction

Dice are commonly used to introduce randomness into games. For example, board games like Monopoly and Catan utilize the sum of two dice to randomly move pieces around a board or to randomly produce resources. Both games are affected by the fact that some sums are more likely to be rolled than others. While both games come with two standard dice there is another pair of dice that could be used.

What follows is an algebraic approach to show that $\{1, 2, 2, 3, 3, 4\}$ and $\{1, 3, 4, 5, 6, 8\}$ are the only pair of non-standard dice with strictly positive integer values that yield the same probability distribution for their sum as the standard pair of dice with faces $\{1, 2, 3, 4, 5, 6\}$.

Table: Probability Distribution for the Sum of Two Standard Dice

Sum	Ways to Roll	Probability
2	1	1/36
3	2	2/36
4	3	3/36
5	4	4/36
6	5	5/36
7	6	6/36
8	5	5/36
9	4	4/36
10	3	3/36
11	2	2/36
12	1	1/36

Method

The Weird Dice can be derived by:

- ▶ Representing the probability distribution for the sum of two dice as the multiplication of two polynomials
- ▶ Factoring the representation
- ▶ Setting constraints
- ▶ Refining constraints
- ▶ Checking the remaining candidates

Setup

Representing the Probability Distribution:

- ▶ Let a standard die be $(x^6 + x^5 + x^4 + x^3 + x^2 + x)$.
- ▶ Represent the probability distribution with $P(x) = (x^6 + x^5 + x^4 + x^3 + x^2 + x)^2 = (x^{12} + 2x^{11} + 3x^{10} + 4x^9 + 5x^8 + 6x^7 + 5x^6 + 4x^5 + 3x^4 + 2x^3 + x^2)$.
- ▶ For two general dice $A(x)$ and $B(x)$ let $A(x)B(x) = (x^{a_1} + x^{a_2} + x^{a_3} + x^{a_4} + x^{a_5} + x^{a_6})(x^{b_1} + x^{b_2} + x^{b_3} + x^{b_4} + x^{b_5} + x^{b_6}) = P(x)$,
for positive integers $a_1, a_2, \dots, a_6$ and $b_1, b_2, \dots, b_6$.

Factoring the Representation:

- ▶ Factor $P(x)$ such that $P(x) = (x^6 + x^5 + x^4 + x^3 + x^2 + x)^2 = (x(x+1)(x^2+x+1)(x^2-x+1))^2 = x^2(x+1)^2(x^2+x+1)^2(x^2-x+1)^2$.
- ▶ Note that $A(x)$ and $B(x)$ must be constructed from these irreducible factors.

Deriving The Weird Dice

Setting Constraints:

- ▶ Then:
 $A(x) = x^{q_A}(x+1)^{r_A}(x^2+x+1)^{s_A}(x^2-x+1)^{t_A}$
 $B(x) = x^{q_B}(x+1)^{r_B}(x^2+x+1)^{s_B}(x^2-x+1)^{t_B}$
for non-negative integers q, r, s, t with $0 \leq q, r, s, t \leq 2$.
- ▶ Each factor must appear twice which requires $q_A + q_B = 2, \quad r_A + r_B = 2,$
 $s_A + s_B = 2, \quad t_A + t_B = 2.$

Refining Constraints:

- ▶ Given:
 $A(x) = (x^{a_1} + x^{a_2} + x^{a_3} + x^{a_4} + x^{a_5} + x^{a_6}) = x^{q_A}(x+1)^{r_A}(x^2+x+1)^{s_A}(x^2-x+1)^{t_A}$.
- ▶ Consider $x = 1$,
 $A(1) = 1^{a_1} + 1^{a_2} + 1^{a_3} + 1^{a_4} + 1^{a_5} + 1^{a_6} = 1^{q_A}(1+1)^{r_A}(1^2+1+1)^{s_A}(1^2-1+1)^{t_A} = 1 + 1 + 1 + 1 + 1 + 1 = 1^{q_A}2^{r_A}3^{s_A}1^{t_A} = 6 = 2^{r_A}3^{s_A},$
which provides that $r_A = 1, s_A = 1$.
- ▶ Consider $x = 0$,
 $A(0) = 0^{a_1} + 0^{a_2} + 0^{a_3} + 0^{a_4} + 0^{a_5} + 0^{a_6} = 0^{q_A}(0+1)^{r_A}(0^2+0+1)^{s_A}(0^2-0+1)^{t_A} = 0 = 0^{q_A}1^{r_A}1^{s_A}1^{t_A} = 0^{q_A},$
which provides that $q_A \neq 0$.

- ▶ Apply the same previous steps to $B(x)$ to obtain $r_B = 1, s_B = 1$, and $q_B \neq 0$.
- ▶ Since $q_A \neq 0, q_B \neq 0$, and $q_A + q_B = 2$, it follows that $q_A = q_B = 1$.

Checking the Remaining Candidates:

- ▶ Check $t = 0$,
 $A(x) = x^4 + 2x^3 + 2x^2 + x \Rightarrow \{1, 2, 2, 3, 3, 4\}.$
- ▶ Check $t = 1$,
 $A(x) = x^6 + x^5 + x^4 + x^3 + x^2 + x \Rightarrow \{1, 2, 3, 4, 5, 6\}.$
- ▶ Check $t = 2$,
 $A(x) = x^8 + x^6 + x^5 + x^4 + x^3 + x \Rightarrow \{1, 3, 4, 5, 6, 8\}.$
- ▶ Since $t_A + t_B = 2$, the possible splits are $(t_A, t_B) = (0, 2), (1, 1), (2, 0)$.

Conclusion

There is one pair of non-standard dice with strictly positive integer faces that share the same probability distribution for their sum as the sum of two standard dice. That pair, named the Sicherman dice, have faces with the values

$$\{1, 2, 2, 3, 3, 4\} \quad \text{and} \quad \{1, 3, 4, 5, 6, 8\}.$$

History

The Weird Dice were discovered by George Sicherman, a retired computer programmer and puzzle inventor, in 1977. George was born in Buffalo, NY, and currently lives in Ocean, NJ. His main hobby is designing puzzle games.

References

- [1] Joseph Gallian. *Weird Dice*. Math Horizons, Vol. 2, No. 3 (February 1995), pp. 30-31, <https://www.jstor.org/stable/25678009>.